**МІНІСТЕРСТВО ОСВІТИ І НАУКИ УКРАЇНИ**

**НАЦІОНАЛЬНИЙ УНІВЕРСИТЕТ “ЛЬВІВСЬКА ПОЛІТЕХНІКА”**

**Кафедра систем штучного інтелекту**




# Лабораторна робота № 3

з дисципліни «Хмарні технології»

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Лозовий Олег

**Викладач:**

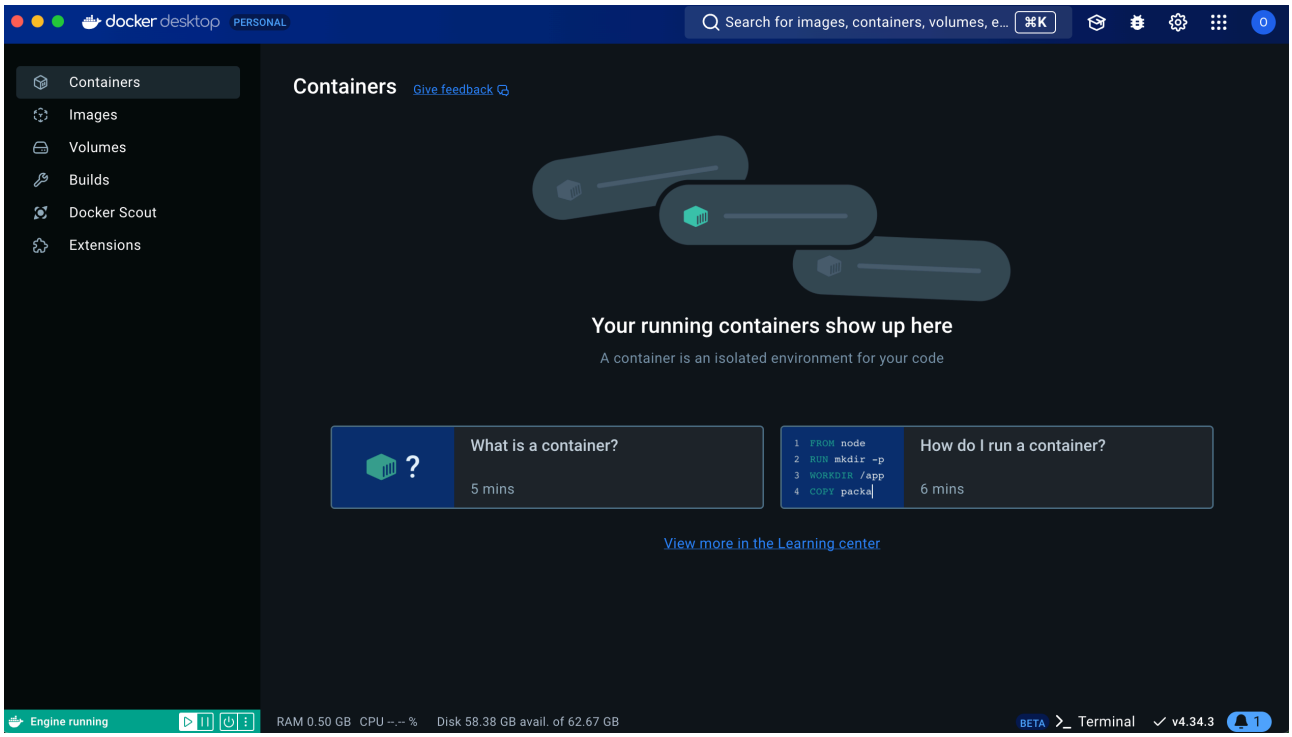
Гасько Р.Т.

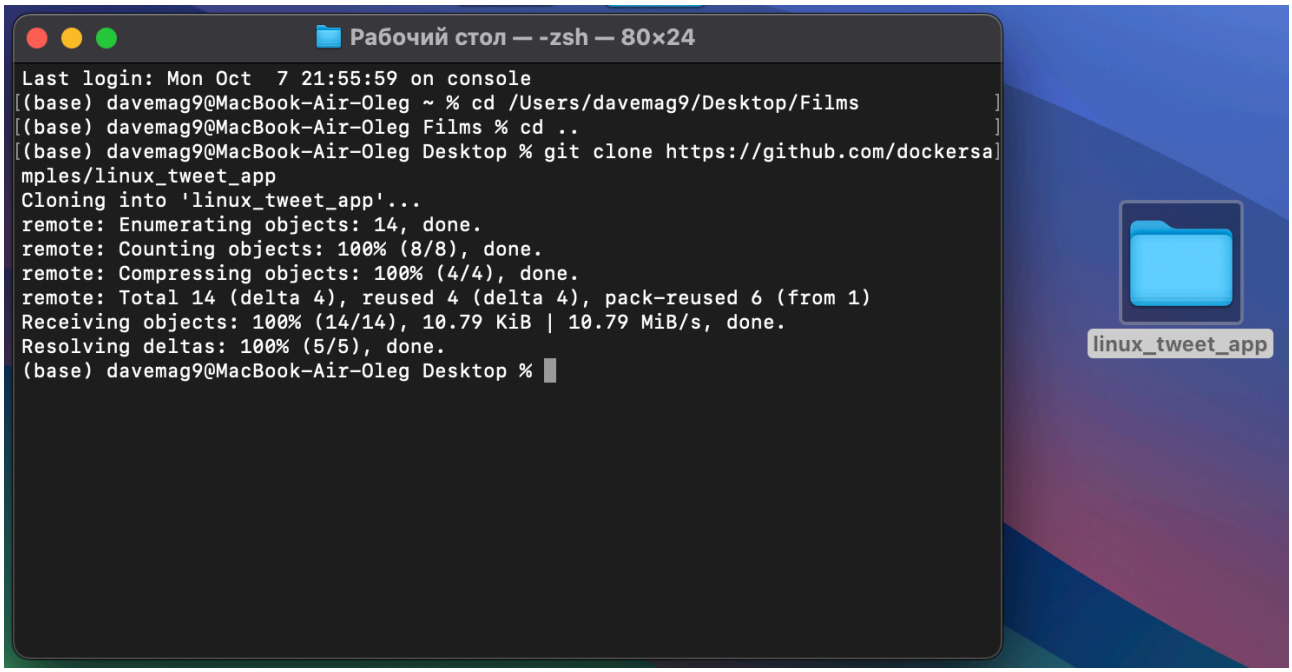
Львів – 2024р.

**Тема:** Containers. Basic operation with Docker.

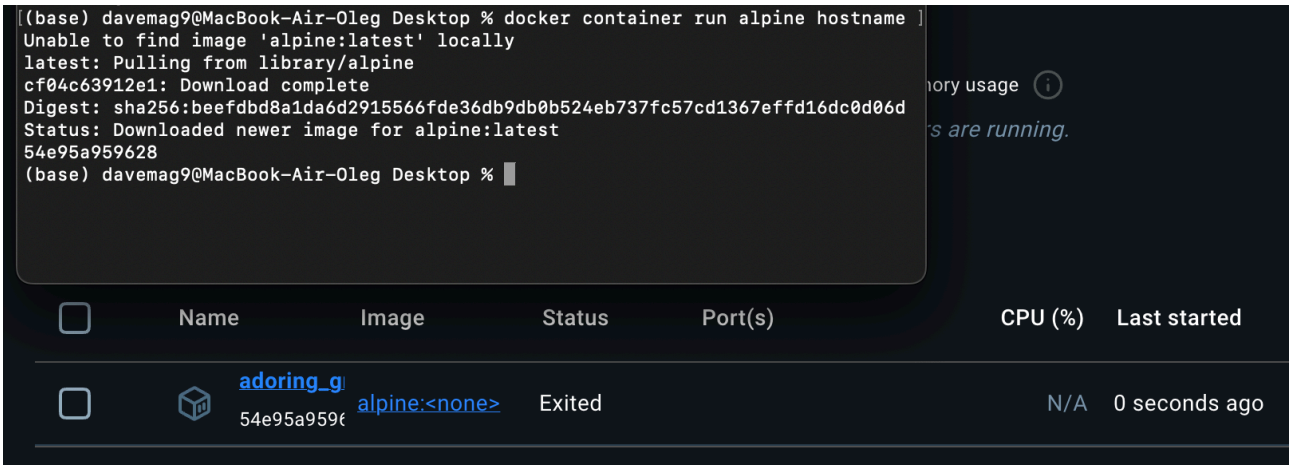
**Хід роботи:**

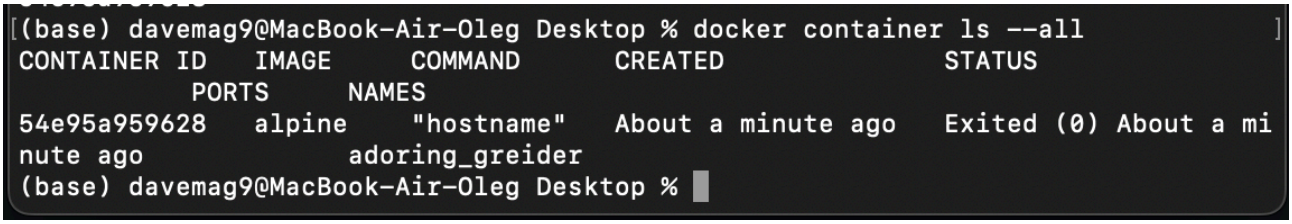
Task 0: Prerequisites



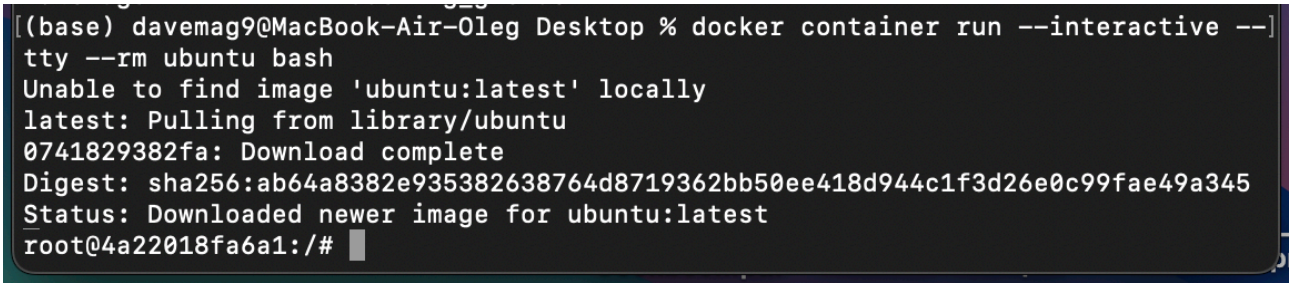



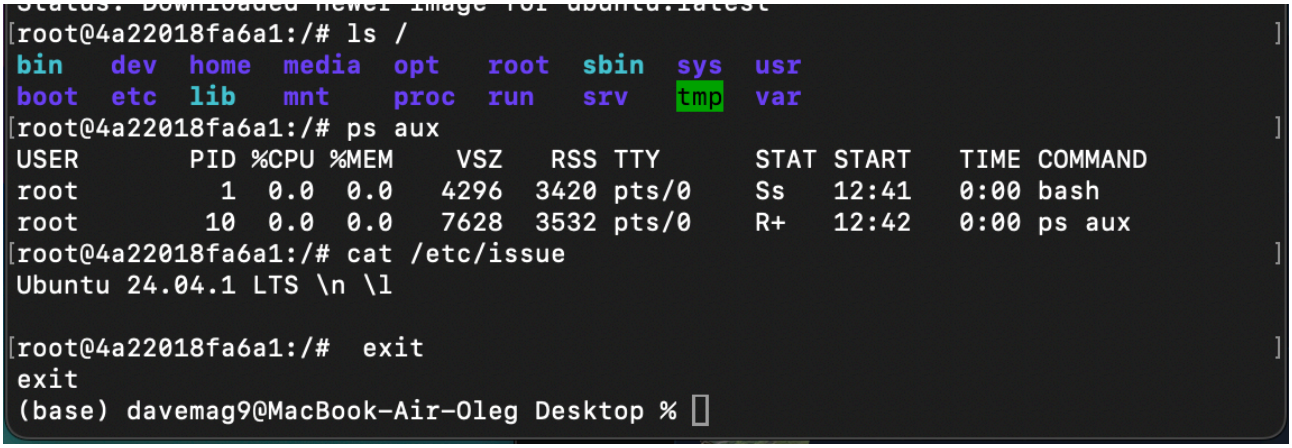
## Task 1: Run some simple Docker containers




Run an interactive Ubuntu container



Run a background MySQL container

```
(base) davemag9@MacBook-Air-Oleg Desktop %  docker container run \
 --detach \
 --name mydb \
 -e MYSQL_ROOT_PASSWORD=my-secret-pw \
 mysql:latest
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
883258893faf: Download complete
08f5c00014df: Download complete
65b13290a890: Download complete
d4f83991d877: Download complete
f88be74802d6: Download complete
8b4274ea61c5: Download complete
9543fc93cec2: Download complete
f8d978a6739e: Download complete
4a0edbf0dd13: Download complete
64da6e9087ad: Download complete
Digest: sha256:92dc869678019f65d761155dacac660a904f6245bfe1b7997da0a73b2bfc68c9
Status: Downloaded newer image for mysql:latest
14383f58bdded2da0cdc46c750c738e3ae5e6886d652eba169284f47c66b607f
```

```
(base) davemag9@MacBook-Air-Oleg Desktop % docker container top mydb
UID                 PID                 PPID                C                   STIME               TTY
          TIME                CMD
999                 744                 725                 0                   12:46               ?
          00:00:00            mysqld
```

```
(base) davemag9@MacBook-Air-Oleg Desktop % docker exec -it mydb \
 mysql --user=root --password=$MYSQL_ROOT_PASSWORD --version
mysql: [Warning] Using a password on the command line interface can be insecure.
mysql  Ver 9.0.1 for Linux on aarch64 (MySQL Community Server - GPL)

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug mydb
    Learn more at https://docs.docker.com/go/debug-cli/
```

```
(base) davemag9@MacBook-Air-Oleg Desktop % docker exec -it mydb sh
sh-5.1#
sh-5.1#
sh-5.1#
sh-5.1# mysql --user=root --password=$MYSQL_ROOT_PASSWORD --version
mysql: [Warning] Using a password on the command line interface can be insecure.
mysql  Ver 9.0.1 for Linux on aarch64 (MySQL Community Server - GPL)
sh-5.1# exit
exit
```

Task 2: Package and run a custom app using Docker

Build a simple website image

```
Last login: Mon Oct 14 15:50:03 on ttys000
(base) davemag9@MacBook-Air-Oleg ~ % cd /Users/davemag9/Desktop/linux_tweet_app

(base) davemag9@MacBook-Air-Oleg linux_tweet_app % cat Dockerfile
FROM nginx:latest

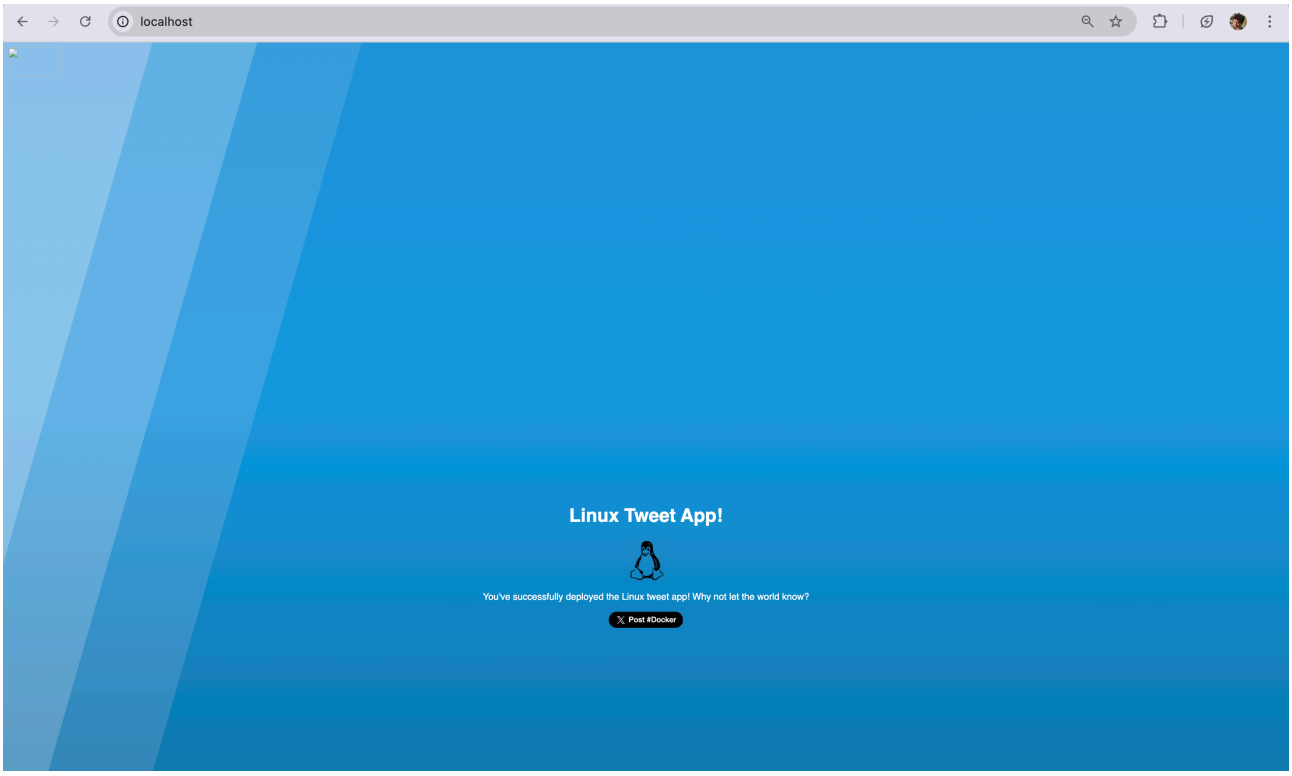
COPY index.html /usr/share/nginx/html
COPY linux.png /usr/share/nginx/html

EXPOSE 80 443

CMD ["nginx", "-g", "daemon off;"]
(base) davemag9@MacBook-Air-Oleg linux_tweet_app % export DOCKERID=olehlozoviy
(base) davemag9@MacBook-Air-Oleg linux_tweet_app % echo $DOCKERID
olehlozoviy
```

```
(base) davemag9@MacBook-Air-Oleg linux_tweet_app % docker image build --tag $DOC
KERID/linux_tweet_app:1.0 .
[+] Building 23.4s (9/9) FINISHED                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                       0.0s
 => => transferring dockerfile: 186B                                       0.0s
 => [internal] load metadata for docker.io/library/nginx:latest            4.9s
 => [auth] library/nginx:pull token for registry-1.docker.io               0.0s
 => [internal] load .dockerignore                                          0.0s
 => => transferring context: 2B                                            0.0s
 => [1/3] FROM docker.io/library/nginx:latest@sha256:d2eb56950b84efe34f9  18.2s
 => => resolve docker.io/library/nginx:latest@sha256:d2eb56950b84efe34f96  0.0s
 => => sha256:66b98279bfb92df7ae2c28fa75e830faa9a7226079e 1.40kB / 1.40kB  0.2s
 => => sha256:e905ecdf6023bc5ecc0c08ad95b3b946340fd389387 1.21kB / 1.21kB  0.4s
 => => sha256:e0917afa5d8b7098a9044df4dbfa166c23ec293e703f132 403B / 403B  0.5s
 => => sha256:e9346e20ffd27cbb75c776445a5d69b2624e840e40733b3 954B / 954B  0.6s
 => => sha256:783efa7200714b88d9642a0b17a3c1b100033df3b8873ec 629B / 629B  0.4s
 => => sha256:c62fa01086040cd1f46a1cca2e6d9757c0516e50 40.42MB / 40.42MB  10.5s
 => => sha256:14c9d9d199323cbf0a4c2347a8af85f2875c1f2c 29.16MB / 29.16MB  16.4s
 => => extracting sha256:14c9d9d199323cbf0a4c2347a8af85f2875c1f2c26a1558f  0.6s
 => => extracting sha256:c62fa01086040cd1f46a1cca2e6d9757c0516e5076b0c47c  0.6s
 => => extracting sha256:783efa7200714b88d9642a0b17a3c1b100033df3b8873ec2  0.0s
 => => extracting sha256:e9346e20ffd27cbb75c776445a5d69b2624e840e40733b3b  0.0s
 => => extracting sha256:e0917afa5d8b7098a9044df4dbfa166c23ec293e703f1324  0.0s
```

```
    View a summary of image vulnerabilities and recommendations →
(base) davemag9@MacBook-Air-Oleg linux_tweet_app % docker containe
 --detach \
 --publish 80:80 \
 --name linux_tweet_app \
 $DOCKERID/linux_tweet_app:1.0
9bf700633b8790a9795e96474a9b714044ca17aebb51c91bf2a2cbaeec4595fa
(base) davemag9@MacBook-Air-Oleg linux_tweet_app %
```




```
(base) davemag9@MacBook-Air-Oleg linux_tweet_app % docker container rm --force linux_tweet_app
linux_tweet_app
(base) davemag9@MacBook-Air-Oleg linux_tweet_app %
```

Task 3: Modify a running website

```
(base) davemag9@MacBook-Air-Oleg linux_tweet_app % docker container run \
 --detach \
 --publish 80:80 \
 --name linux_tweet_app \
 --mount type=bind,source="$(pwd)",target=/usr/share/nginx/html \
 $DOCKERID/linux_tweet_app:1.0
99cdca897d1999012cf359dc6c89d26b418d182bc20151fc7e0ff162e4c857c3
(base) davemag9@MacBook-Air-Oleg linux_tweet_app %
```

Start our web app with a bind mount

```
(base) davemag9@MacBook-Air-Oleg linux_tweet_app % cp index-new.html index.html
(base) davemag9@MacBook-Air-Oleg linux_tweet_app % docker rm --force linux_tweet_app
linux_tweet_app
(base) davemag9@MacBook-Air-Oleg linux_tweet_app % docker container run \
 --detach \
 --publish 80:80 \
 --name linux_tweet_app \
 $DOCKERID/linux_tweet_app:1.0
469da6452b9600fea2fdcf02a6495cd4b877c6204d7ce606fa5471674a8ce06c
(base) davemag9@MacBook-Air-Oleg linux_tweet_app % docker rm --force linux_tweet_app
linux_tweet_app
(base) davemag9@MacBook-Air-Oleg linux_tweet_app %
```

Update the image

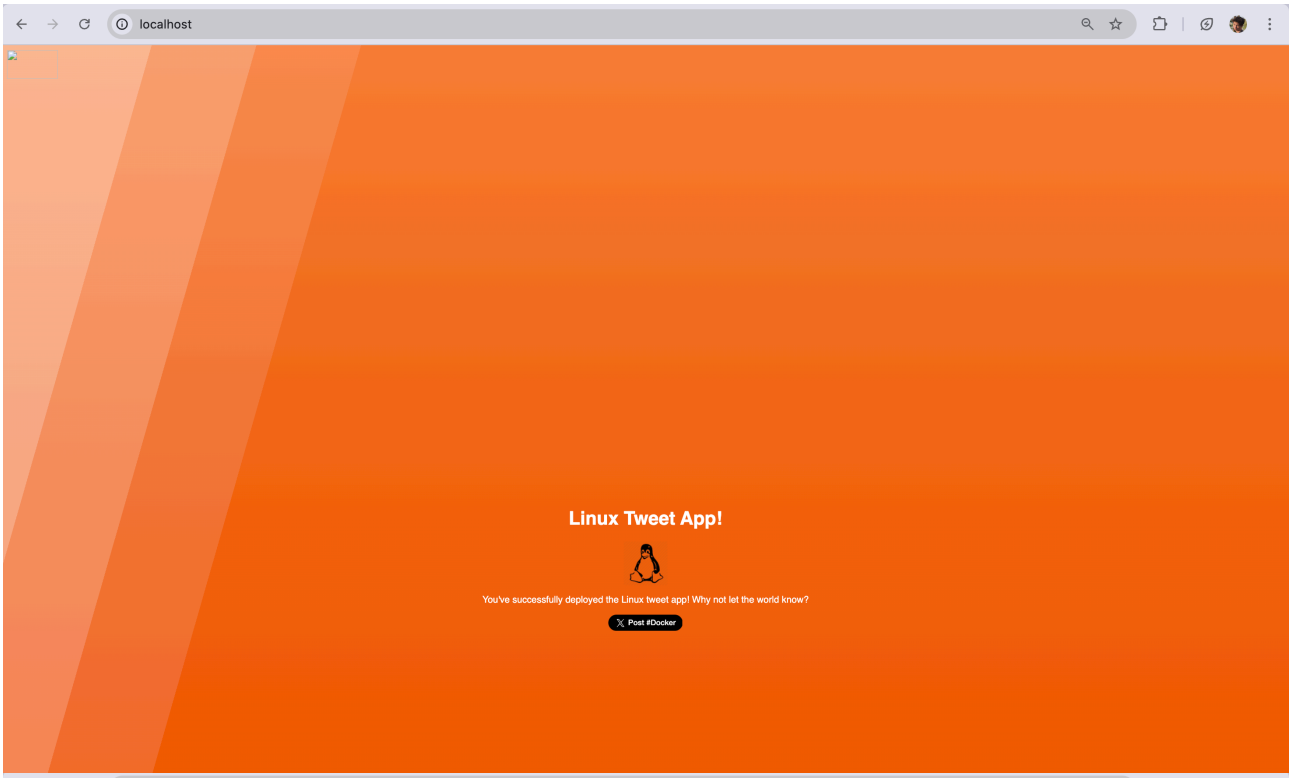
```
(base) davemag9@MacBook-Air-Oleg linux_tweet_app % docker image build --tag $DOCKERID/linux_tweet_app:2.0 .

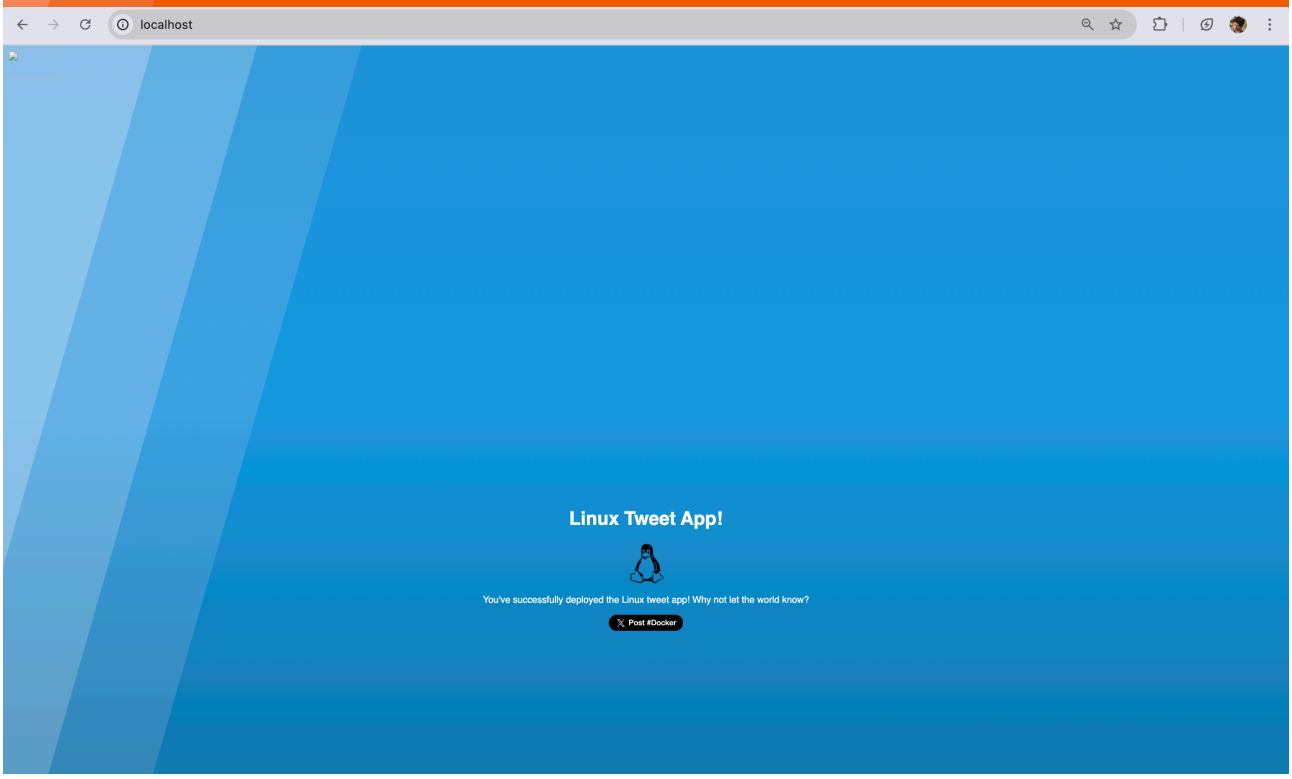
[+] Building 1.8s (9/9) FINISHED                                                docker:desktop-linux
 => [internal] load build definition from Dockerfile                                          0.0s
 => => transferring dockerfile: 186B                                                          0.0s
 => [internal] load metadata for docker.io/library/nginx:latest                               1.4s
 => [auth] library/nginx:pull token for registry-1.docker.io                                  0.0s
 => [internal] load .dockerignore                                                             0.0s
 => => transferring context: 2B                                                               0.0s
 => CACHED [1/3] FROM docker.io/library/nginx:latest@sha256:d2eb56950b84efe34f966a2b92efb1a1a2ea53e7  0.0s
 => => resolve docker.io/library/nginx:latest@sha256:d2eb56950b84efe34f966a2b92efb1a1a2ea53e7e93b94c  0.0s
 => [internal] load build context                                                             0.0s
 => => transferring context: 1.74kB                                                           0.0s
 => [2/3] COPY index.html /usr/share/nginx/html                                               0.0s
 => [3/3] COPY linux.png /usr/share/nginx/html                                                0.0s
 => exporting to image                                                                        0.2s
 => => exporting layers                                                                       0.1s
```

```
(base) davemag9@MacBook-Air-Oleg linux_tweet_app % docker image ls
REPOSITORY                    TAG       IMAGE ID       CREATED              SIZE
olehlozoviy/linux_tweet_app   2.0       6cf14267071e   About a minute ago   281MB
olehlozoviy/linux_tweet_app   1.0       60362ac14ae9   17 minutes ago       281MB
ubuntu                        latest    ab64a8382e93   4 days ago           139MB
alpine                        latest    beefdbd8a1da   5 weeks ago          13.6MB
mysql                         latest    92dc86967801   2 months ago         823MB
```

Test the new version

```
(base) davemag9@MacBook-Air-Oleg linux_tweet_app % docker container run \
 --detach \
 --publish 80:80 \
 --name linux_tweet_app \
 $DOCKERID/linux_tweet_app:2.0
dedafe01219a21547517e5fa718adc9484bd85cba022fa15c9b43fb1a88d6c76
(base) davemag9@MacBook-Air-Oleg linux_tweet_app %  docker container run \
 --detach \
 --publish 8080:80 \
 --name old_linux_tweet_app \
 $DOCKERID/linux_tweet_app:1.0
b6f95abd921fc30e7b84ffe062820f45ff3277ca2ac3f9a7ca28f4b522103111
(base) davemag9@MacBook-Air-Oleg linux_tweet_app %  docker image ls -f reference="$DOCKERID/*"
REPOSITORY                    TAG       IMAGE ID       CREATED          SIZE
olehlozoviy/linux_tweet_app   2.0       6cf14267071e   2 minutes ago    281MB
olehlozoviy/linux_tweet_app   1.0       60362ac14ae9   17 minutes ago   281MB
```



Push your images to Docker Hub

```
(base) davemag9@MacBook-Air-Oleg linux_tweet_app % docker login
Authenticating with existing credentials...
Login Succeeded
(base) davemag9@MacBook-Air-Oleg linux_tweet_app %  docker image push $DOCKERID/linux_tweet_app:1.0
The push refers to repository [docker.io/olehlozoviy/linux_tweet_app]
e0917afa5d8b: Pushed
e905ecdf6023: Pushed
66b98279bfb9: Pushed
783efa720071: Pushed
3b2c6ca13ee8: Pushed
1b26041e3e06: Pushed
14c9d9d19932: Pushed
0bc020fbb561: Pushed
e9346e20ffd2: Pushed
c62fa0108604: Pushed
1.0: digest: sha256:60362ac14ae94d648b82f7b3163c855dd2cda85e16f272c435cef685a1f8b045 size: 856
(base) davemag9@MacBook-Air-Oleg linux_tweet_app % docker image push $DOCKERID/linux_tweet_app:2.0
The push refers to repository [docker.io/olehlozoviy/linux_tweet_app]
f1c55b5cb03a: Pushed
c62fa0108604: Layer already exists
e905ecdf6023: Layer already exists
4c727693a59f: Pushed
e9346e20ffd2: Layer already exists
783efa720071: Layer already exists
f6f02d096dba: Pushed
14c9d9d19932: Layer already exists
66b98279bfb9: Layer already exists
e0917afa5d8b: Layer already exists
2.0: digest: sha256:6cf14267071e96d12386153d82d1effa3a0937c9ae64059189b5555b99be9806 size: 856
```




Висновок: під час виконання даної лабораторної роботи, я навчився користуватись Doker.